



Macro Exposure

Field manual · Canon EOS R5 with the Laowa macro set

This app answers the four questions that manual macro work keeps asking and that nothing on the camera will tell you: what aperture the sensor is *actually* seeing, how thin the plane of focus has become, how far to advance the rail, and how much flash power the last change just cost you.

It runs entirely on the phone. No network, no GPS, no permissions, no account. Once the page has loaded it works in a cellar or a swamp with the phone in airplane mode.

Where it lives	What it needs
executivedesign.github.io/valley-view-trails/macro/	Any phone browser. Nothing else.

Install it properly. Open the link in Safari, then Share > Add to Home Screen. That gives you an icon, full screen, and a copy cached on the phone so it opens without a signal. Settings are remembered between sessions.

Everything in this manual describes the app as built. The formulas are given in full at the end, so nothing here has to be taken on trust.

The one idea

At macro magnifications the number on the aperture ring is close to fiction. Extending a lens to focus close moves the aperture further from the sensor, and the light spreads out over that extra distance. What reaches the sensor follows the **effective aperture**:

$$N_{\text{eff}} = N \times (1 + m/P)$$

where N is the marked f-number, m is magnification, and P is the lens's pupil magnification. The light lost, in stops, is $2 \times \log_2(1 + m/P)$.

Wide open, across your kit:

Lens	Magnification	Marked	Effective	Light lost
15mm f/4 Shift	1:1	f/4	f/8	2.0 stops
24mm f/14 Probe	2:1	f/14	f/42	3.2 stops
25mm Ultra	2.5x	f/2.8	f/9.8	3.6 stops
25mm Ultra	5x	f/2.8	f/16.8	5.2 stops
100mm f/2.8 APO	2:1	f/2.8	f/7.3	2.7 stops
180mm f/4.5 APO	1.5x	f/4.5	f/11.2	2.6 stops
Aurogon, NA 0.5	20x	—	f/21	—
Aurogon, NA 0.5	50x	—	f/51	—

The probe is the clearest case: f/14 is as wide as it opens, and at 2:1 that behaves like f/42. Four stops darker than the barrel says. This is why open shade on that lens needs ISO 3200 and it is not the meter being wrong.

What the light loss does and does not mean

The R5 meters off the sensor. With a fully manual lens the meter is reading the light that *actually arrived*, so the loss is already in what the camera shows you. **It is not a correction to dial in.** Adding it again would overexpose by several stops.

It matters for three other things, all of which the app handles:

- **Flash power** — manual flash cannot see through the lens, so the correction is yours to make.
- **ISO headroom** — knowing you are shooting at f/42 explains why ISO is climbing.
- **Diffraction and depth of field** — both follow the effective aperture, never the marked one.

Why the 100mm is different

Most lenses are treated as symmetric, $P = 1$. The 100mm f/2.8 APO is not, and Laowa publishes its real losses: 1.7 stops at 1:1 and 2.7 at 2x. Those two figures fix $P = 1.29$, which reproduces both to within a twentieth of a stop and predicts everything in between. So that lens is modelled from measured data rather than theory, and it loses about a third of a stop less than the naive formula would claim.

The Shoot tab

Set three things: lens, magnification, aperture. Everything below updates.

Lens

Six buttons. Choosing one loads that lens's real limits, so the app will not offer you f/4 on the probe or 3x on the 180mm — and magnification clamps into range if the new lens cannot reach where you were.

Magnification

A slider with a number field beside it. On the Aurogon it becomes four tube buttons, since that lens only does 10x, 20x, 35x and 50x.

Aperture

Stepped in thirds through the range that lens actually has. On the Aurogon it becomes NA instead — eight detents from 0.5 to 0.15 — because a microscope objective is specified by numerical aperture. The app converts it with the working f-number, $(1 + m) / 2NA$.

What you get back

Reading	What it tells you
Effective aperture	The headline figure, in the large type. What the sensor sees.
Light lost	How many stops separate that from the barrel marking.
Working distance	Front of lens to subject, at Laowa's nearest documented magnification — the app names which one it used.
Airy disk	The diffraction blur diameter, in microns, against your 4.4 μm pixels.
Scene light	The ambient EV at ISO 100 implied by your ISO and shutter.
Handheld floor	The slowest shutter worth attempting, $1/(\text{focal} \times (1 + m))$. Shake is magnified along with the subject.

Diffraction

Reported in three bands, not as a single threshold, because diffraction is not a cliff — it costs resolution steadily long before anyone would call a print soft.

- **Not diffraction limited** — below about f/6.5 effective.
- **Softening at 100%, invisible in a print** — roughly f/6.5 to f/22.
- **Visible in print too** — past about f/22 effective.

Every lens in your kit is at or past the first threshold *wide open* at working magnification. That is not a fault and it is not avoidable. On this kit the question is never whether you are diffracted, only which compromise you would rather have — which is exactly the trade the Stack tab lets you price.

The Stack tab

Depth of field at macro distances does not behave like the depth of field you are used to. It follows:

$$\text{DoF} = 2 \times N_{\text{eff}} \times c / m^2$$

Note the m squared. Doubling magnification quarters your depth of field, and no aperture on any of these lenses can buy that back.

Which circle of confusion

The toggle offers two, and the choice is not cosmetic — it changes the answer roughly threefold.

- **Print, 0.030 mm** — the traditional standard. Right if the picture ends up on a wall or a screen at sensible size.
- **Pixel level, 0.009 mm** — about two pixels on your 45 MP sensor. The honest figure if you inspect at 100%, and the one to stack by if you want it critically sharp everywhere.

Rail step and frame count

Give it the depth of subject you need covered and it returns the step and the number of frames. Step is depth of field less the overlap, defaulting to 30%.

The overlap is not padding. Slices that merely touch leave visible banding where the sharp zones meet, because the edge of a depth of field is a gradual thing rather than a line. Thirty percent is the usual insurance; tighten it if a stack shows seams.

A worked example

25mm Ultra at 5x, wide open at f/2.8, stacking a 3 mm subject:

	Print (0.030 mm)	Pixel level (0.009 mm)
Effective aperture	f/16.8	f/16.8
Depth of field	40 μm	12 μm
Rail step at 30%	28 μm	8.5 μm
Frames for 3 mm	108	356

Same subject, same settings, three times the work for the stricter standard. That is the decision the toggle is there to let you make deliberately rather than by accident.

The app warns when the step falls below about 5 μm , which is finer than most manual rails resolve — on the Aurogon that happens immediately, and at 20x the step is around 2 μm . That is motorised controller territory, not thumb and micrometer.

The Flash tab

Five of your six lenses have no electronic contacts. The camera therefore has no aperture to work from, TTL cannot compute a correct exposure, and you are on manual power whether you wanted to be or not.

Guide numbers do not rescue you either: they are quoted in metres, and at a 20 mm working distance the arithmetic has left the range where it means anything.

So the app works from a baseline

Get one frame right by eye. Then open the Flash tab, set **Power then** to the power you used, and press **Save current as baseline**. It records the magnification, the effective aperture and the ISO alongside it.

From that point on, change anything you like. **Power now** tells you what to set, and by how many stops the requirement moved:

$$\Delta \text{stops} = 2 \times \log_2(N_{\text{now}} / N_{\text{base}}) - \log_2(\text{ISO}_{\text{now}} / \text{ISO}_{\text{base}})$$

Power is displayed the way the Godox units display it — a fraction plus thirds, so “1/4 +1/3” is one third of a stop above quarter power.

Example

Baseline: 100mm APO at 1:1, marked f/8, ISO 400, MF-12 twin at 1/16. Now you move to 2:1 and leave everything else alone. Effective aperture goes from f/14.2 to f/20.4 — just over a stop — so the app calls for 1/8. Nothing else on the camera changed, and nothing on the camera would have told you.

Shutter speed is deliberately not in that formula

This is the part that trips people up, so it is worth being explicit. **Shutter speed does not affect flash exposure.** The pop lasts a fraction of a millisecond; below sync speed the whole of it lands in the frame whether you are at 1/60 or 1/200.

Shutter controls your *ambient* only. That is a feature: drag it to bring the background up or shut it down, without touching how the flash renders your subject.

Sync is a hard wall. The R5 syncs at 1/200 with the mechanical shutter and 1/250 with electronic first curtain, and will not fire flash at all on the fully electronic shutter. Go past it and a black band crosses the frame. The app refuses to hand you a setting beyond that rather than letting you find out later.

Continuous light

Selecting the LED option turns the power maths off, because there is nothing to calculate — continuous light meters normally and the shutter is back in play. What the effective aperture still costs you is output, and at f/17 or f/42 the Lume Cubes and RM03s will be working hard. Expect long exposures on a rail.

Lens quick reference

Lens	Mag	Aperture	Working dist.	Contacts
15mm f/4 Shift	to 1:1	f/4–f/32	5 mm at 1:1	None
24mm f/14 Probe	to 2:1	f/14–f/40	20 mm at 2:1	None
25mm Ultra	2.5–5x	f/2.8–f/16	40–45 mm	None
100mm f/2.8 APO	to 2:1	f/2.8–f/22	71 mm at 2x	Full
180mm f/4.5 APO	to 1.5x	f/4.5–f/32	147 mm at 1.5x	None
Aurogon	10–50x	NA 0.5–0.15	20 mm, fixed	None

Working distance is front of lens to subject — the one you can judge. Minimum focus distance, quoted from the sensor plane, is always longer and less useful in the field. The app shows both.

The 100mm is the only lens here that meters, autoexposes, records EXIF and can drive TTL. When something behaves unexpectedly, check which lens you are on first.

Per-lens notes live in the app

The bottom of the Shoot tab carries that lens's handling notes — which ones shadow the subject, which need the hood removed, where IBIS must be switched off, which need the focal length entered by hand. They are drawn from your own settings reference so the app and your notes cannot disagree.

Warnings you may see

- **Past your ISO ceiling** — above 6400, the limit you set.
- **Slower than the handheld floor** — fine on a rail. Handheld, let the flash duration be your real shutter.
- **Aurogon** — electronic shutter only, IBIS off, remote release, hands off the camera entirely.
- **No electronic contacts** — on five of the six lenses nothing about the exposure reaches EXIF. Write it down if it matters.

Troubleshooting

The effective aperture looks alarming. It is meant to. f/42 on the probe is correct, not a bug — it is the reason that lens needs so much light.

It says I am diffracted wide open. On this kit you usually are. Choose for depth of field instead and accept the softening; a stacked, slightly diffracted frame beats a sharp one with nothing in focus.

The frame count is enormous. Switch the circle of confusion to print, or stop down and take the diffraction. Both are legitimate, and the app shows what each costs.

Flash power says beyond full. You have run out of light. Raise ISO, open up, or move the heads closer — halving the distance is worth two stops.

Nothing appears in EXIF. Expected on five of six lenses. There are no contacts to carry it.

The formulas, in full

Nothing in the app is hidden. Every number on the screen comes from one of these.

Effective aperture

$$N_{\text{eff}} = N \times (1 + m/P)$$

$P = 1$ for a symmetric lens, 1.29 for the 100mm APO, fitted to Laowa's published losses.

Light lost

$$\text{stops} = 2 \times \log_2(1 + m/P)$$

Aurogon working f-number

$$N_{\text{eff}} = (1 + m) / (2 \times \text{NA})$$

Depth of field

$$\text{DoF} = 2 \times N_{\text{eff}} \times c / m^2$$

$c = 0.030$ mm for print, 0.009 mm at pixel level on the R5's 45 MP sensor.

Rail step and frames

$$\text{step} = \text{DoF} \times (1 - \text{overlap})$$

$$\text{frames} = \text{ceiling}(\text{depth} / \text{step}) + 1$$

Diffraction

$$\text{Airy diameter} = 2.44 \times \lambda \times N_{\text{eff}}, \quad \lambda = 550 \text{ nm}$$

Compared against the R5 pixel pitch of 4.39 μm , and against the print circle of confusion for the second threshold.

Flash power change

$$\Delta \text{stops} = 2 \times \log_2(N_{\text{now}}/N_{\text{base}}) - \log_2(\text{ISO}_{\text{now}}/\text{ISO}_{\text{base}})$$

Handheld floor

$$\text{shutter} = 1 / (\text{focal length} \times (1 + m))$$

Scene light

$$\text{EV}_{100} = \log_2(N_{\text{eff}}^2 / t) - \log_2(\text{ISO} / 100)$$

Camera figures used throughout: Canon EOS R5, 45 MP full frame, 4.39 μm pixel pitch, flash sync 1/200 mechanical and 1/250 with electronic first curtain, shutter 30 s to 1/8000, ISO 100–51200. Lens specifications are taken from the owner's Laowa and R5 settings reference.